GRADE 10 / MYP 5 • FORCE & MOTION

# Toy-car investigation: analyse motion

(Pre-Crit B/C assessment) • Use Desmos, create velocity–time data and review an investigation plan.

TODAY'S GOALS

# From data to explanations

- Plot your own x–t data in Desmos.
- Compare manual and digital gradient estimates.
- Calculate interval-average velocities and plot them against time.
- Explain how velocity changes and evaluate an experimental plan.

DESMOS: START WITH YOUR DATA

# Create a position–time plot

- Open https://www.desmos.com/calculator.
- Press + and add a Table.
- Enter mean elapsed time (s) in $x_1$; position (m) in $y_1$.
- Adjust the view so all the data can be seen.

LOOK FOR THE TREND

# Fitting and gradients

- Show the measured points first.
- Add a smooth curve that represents the pattern.
- Compare an early gradient with a late gradient.
- Keep units: each x–t gradient is in m/s.

EQUATIONS

# Velocity from neighbouring measurements

$$v_{interval} = \Delta x / \Delta t$$

For each consecutive 5 cm interval: Δx = +0.05 m.

Example: Δt = 0.25 s → vinterval = +0.20 m/s.

EQUATIONS

# When did that velocity occur?

$$t_{mid} = (t_{start} + t_{end}) / 2$$

Associate each interval-average velocity with its MIDPOINT time.

Example: 0.60 to 0.85 s → $t_{mid}$ = 0.725 s.

BUILD THE VELOCITY–TIME PLOT

# A second Desmos table

- First column: midpoint time, $t_{mid}$ (s).
- Second column: interval-average velocity, v (m/s).
- Draw a suitable best-fit trend.
- What does its slope tell you?

TWO DIFFERENT COMPARISONS

# Distance or time?

## CHANGE PER SECOND

$\Delta v/\Delta t \rightarrow$ acceleration

Unit: $m/s^2$

## CHANGE PER CENTIMETRE

$\Delta v/\Delta x \rightarrow$ how velocity changes with position.

Unit: $s^{-1}$ (x in metres).

In physics, we always describe acceleration as change in velocity per unit time, rather than per unit distance.

INTERPRET YOUR OWN RESULTS

# Use evidence, not guesses

- Does the x–t curve become steeper?
- Does velocity increase with time?
- Is the v–t trend roughly straight?
- What limitations did you notice while measuring?

CRITERION B PRACTICE

# Read a provided investigation plan

- A1 Research question and explanation
- A2 Hypothesis and scientific reasoning
- A3 Variables
- A4 Method, apparatus and data collection

CRITIQUE THE PLAN

# Use the supplied rubric

- For each section, identify what works well.
- Identify missing or unclear details.
- Suggest specific, justified improvements.
- Support your judgement with the rubric.

HOMEWORK

# Assessment practice

- Complete the four-part Criterion B critique sheet.
- Be ready to explain your strongest proposed improvement.
- Keep your graphs and data for later discussion.

LESSON SUMMARY

# What you should now understand

- The slope of x–t shows velocity.
- Interval Δx/Δt data can produce an approximate v–t graph.
- The slope of v–t shows acceleration.
- A strong scientific plan specifies measurements and controls clearly.